

PART A

$$\begin{aligned}
 1 \text{ a) } \lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x} &= \lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x} \times \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} \\
 &= \lim_{x \rightarrow 0} \frac{x + 4 - 4}{x(\sqrt{x+4} + 2)} \\
 &= \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x+4} + 2)} \\
 &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+4} + 2} \\
 &= \frac{1}{\sqrt{0+4} + 2} \\
 &= \frac{1}{4}
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } \lim_{x \rightarrow -\infty} \frac{2x}{\sqrt{x^2} - 16} &= \lim_{x \rightarrow -\infty} \frac{\frac{2x}{x}}{\frac{\sqrt{x^2} - 16}{x}} \\
 &= \lim_{x \rightarrow -\infty} \frac{2}{\sqrt{1 - \frac{16}{x^2}}} \\
 &= \frac{2}{-\sqrt{1 - 0}} \\
 &= -2
 \end{aligned}$$

$$2 \text{ a) } f(x) = x^2 + 2x$$

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{[(x+h)^2 + 2(x+h)] - (x^2 + 2x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{x^2 + 2hx + h^2 + 2x + 2h - x^2 - 2x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2hx + h^2 + 2h}{h}
 \end{aligned}$$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{h(2x + h + 2)}{h} \\
 &= \lim_{h \rightarrow 0} 2x + h + 2 \\
 &= 2x + 0 + 2 \\
 &= 2x + 2
 \end{aligned}$$

b i) $y = \sin 4x + \ln(x^2 + 2)$

$$\frac{dy}{dx} = 4\cos 4x + \frac{2x}{x^2 + 2}$$

ii) $y = e^{2x} \sec(3x + 1)$

$$\begin{aligned}
 \frac{dy}{dx} &= e^{2x} (3\sec(3x + 1) \tan(3x + 1)) + \sec(3x + 1)(2e^{2x}) \\
 &= e^{2x} \sec(3x + 1)(3\tan(3x + 1) + 2)
 \end{aligned}$$

3 a) $f(x) = 3x^4 - 4x^3 + 3$

$$f'(x) = 12x^3 - 12x^2$$

$$f''(x) = 36x^2 - 24x$$

when $f''(x) = 0$

$$36x^2 - 24x = 0$$

$$12x(3x - 2) = 0$$

$$x = 0, \quad x = \frac{2}{3}$$

x	$(-\infty, 0)$	$\left(0, \frac{2}{3}\right)$	$\left(\frac{2}{3}, \infty\right)$
$f''(x)$	+	-	+

f is concave up on $(-\infty, 0) \cup \left(\frac{2}{3}, \infty\right)$.

f is concave down on $\left(0, \frac{2}{3}\right)$.

when $x = 0$, $f(0) = 3(0)^4 - 4(0)^3 + 3 = 3$

when $x = \frac{2}{3}$, $f\left(\frac{2}{3}\right) = 3\left(\frac{2}{3}\right)^4 - 4\left(\frac{2}{3}\right)^3 + 3 = \frac{65}{27}$

The inflection points are $(0, 3)$ and $\left(\frac{2}{3}, \frac{65}{27}\right)$

PART B

$$\begin{aligned}
 1. \quad \frac{\sqrt{3} + ai}{1 - \sqrt{3}i} &= \frac{\sqrt{3} + ai}{1 - \sqrt{3}i} \times \frac{1 + \sqrt{3}i}{1 + \sqrt{3}i} \\
 &= \frac{\sqrt{3} + 3i + ai + \sqrt{3}ai^2}{1 - 3i^2} \\
 &= \frac{\sqrt{3} + 3i + ai + \sqrt{3}a(-1)}{1 - 3(-1)} \\
 &= \frac{(\sqrt{3} - \sqrt{3}a) + (a + 3)i}{4}
 \end{aligned}$$

For real number, $a + 3 = 0$

$$a = -3$$

$$\begin{aligned}
 \frac{\sqrt{3} + ai}{1 - \sqrt{3}i} &= \frac{\sqrt{3} - \sqrt{3}(-3)}{4} \\
 &= \sqrt{3}
 \end{aligned}$$

$$2 \text{ a) } \sqrt{x+22} + \sqrt{x+6} = 8$$

$$\sqrt{x+22} = 8 - \sqrt{x+6}$$

$$x + 22 = 64 - 16\sqrt{x+6} + x + 6$$

$$16\sqrt{x+6} = 48$$

$$\sqrt{x+6} = 3$$

$$x + 6 = 9$$

$$x = 3$$

b) $\frac{4}{x+1} < x-2$

$$\frac{4}{x+1} - (x-2) < 0$$

$$\frac{4 - (x^2 - x - 2)}{x+1} < 0$$

$$\frac{-x^2 + x + 6}{x+1} < 0$$

$$\frac{x^2 - x - 6}{x+1} > 0$$

$$\frac{(x-3)(x+2)}{x+1} > 0$$

x	$(-\infty, -2)$	$(-2, -1)$	$(-1, 3)$	$(3, \infty)$
$x-3$	-	-	-	+
$x+2$	-	+	+	+
$x+1$	-	-	+	+
$\frac{(x-3)(x+2)}{x+1}$	-	+	-	+

Solution interval is $(-2, -1) \cup (3, \infty)$

3 a) $f(g(x)) = x^2 + 1$

$$f(x+3) = x^2 + 1$$

Let $y = x+3$

$$x = y-3$$

$$f(y) = (y-3)^2 + 1$$

$$= y^2 - 6y + 10$$

$$f(x) = x^2 - 6x + 10$$

b) Let $y = g^{-1}(x)$

$$g(y) = x$$

$$\frac{2y+3}{y-4} = x$$

$$2y+3 = xy-4x$$

$$4x+3 = xy-2y$$

$$4x+3 = y(x-2)$$

$$y = \frac{4x+3}{x-2}$$

$$g^{-1}(x) = \frac{4x+3}{x-2}$$

$$(f \circ g^{-1})(a) = -10$$

$$f\left(\frac{4a+3}{a-2}\right) = -10$$

$$5 - \left(\frac{4a+3}{a-2}\right) = -10$$

$$15 = \frac{4a+3}{a-2}$$

$$15a - 30 = 4a + 3$$

$$11a = 33$$

$$a = 3$$

4 a) $f(x) = e^{3x} + 2$

$$f(x_1) = f(x_2)$$

$$e^{3x_1} + 2 = e^{3x_2} + 2$$

$$e^{3x_1} = e^{3x_2}$$

$$3x_1 = 3x_2$$

$$x_1 = x_2$$

f is a one-to-one function

b) $D_f = (-\infty, \infty)$

$$e^{3x} > 0$$

$$e^{3x} + 2 > 2$$

$$f(x) > 2$$

$$R_f = (2, \infty)$$

c) Let $y = f^{-1}(x)$

$$f(y) = x$$

$$e^{3y} + 2 = x$$

$$e^{3y} = x - 2$$

$$3y = \ln(x - 2)$$

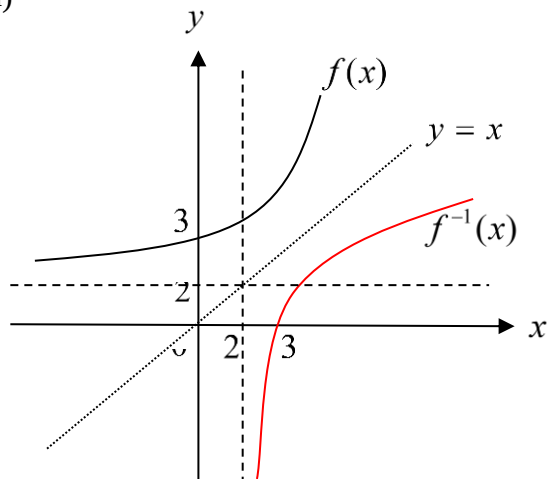
$$y = \frac{1}{3} \ln(x - 2)$$

$$f^{-1}(x) = \frac{1}{3} \ln(x - 2)$$

$$D_{f^{-1}} = (2, \infty)$$

$$R_{f^{-1}} = (-\infty, \infty)$$

d)



$$\begin{aligned}
 5 \text{ a i) } \lim_{x \rightarrow 3} f(x) &= \lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3} \\
 &= \lim_{x \rightarrow 3} \frac{(x - 3)(x^2 + 3x + 9)}{x - 3} \\
 &= \lim_{x \rightarrow 3} x^2 + 3x + 9
 \end{aligned}$$

$$= (3)^2 + 3(3) + 9$$

$$= 27$$

$$\text{ii) } f(3) = 30$$

$$\lim_{x \rightarrow 3} f(x) \neq f(3)$$

f is discontinuous at $x = 3$

b) g is continuous at $x = -2$

$$\lim_{x \rightarrow -2} g(x) \text{ exist}$$

$$\lim_{x \rightarrow -2^-} g(x) = \lim_{x \rightarrow -2^+} g(x)$$

$$\lim_{x \rightarrow -2^-} Ax + B = \lim_{x \rightarrow -2^+} -x^2 + Ax - B$$

$$A(-2) + B = -(-2)^2 + A(-2) - B$$

$$-2A + B = -4 - 2A - B$$

$$2B = -4$$

$$B = -2$$

g is continuous at $x = 2$

$$\lim_{x \rightarrow 2} g(x) \text{ exist}$$

$$\lim_{x \rightarrow 2^-} g(x) = \lim_{x \rightarrow 2^+} g(x)$$

$$\lim_{x \rightarrow 2^-} -x^2 + Ax - B = \lim_{x \rightarrow 2^+} 4$$

$$-(2)^2 + A(2) - (-2) = 4$$

$$-4 + 2A + 2 = 4$$

$$2A = 6$$

$$A = 3$$

c) Let $\sqrt{x^2 - 16} = 0$

$$(x + 4)(x - 4) = 0$$

$$x = -4, \quad x = 4$$

$$\lim_{x \rightarrow -4^-} \frac{2x}{\sqrt{x^2 - 16}} = -\infty$$

$$\lim_{x \rightarrow 4^+} \frac{2x}{\sqrt{x^2 - 16}} = \infty$$

The lines $x = -4$ and $x = 4$ are vertical asymptotes of h .

6 a) For $x \leq 1$, $f'(x) = 4x$

$$f'(1^-) = 4(1) = 4$$

$$f'(1^+) = 5$$

$$f'(1^-) \neq f'(1^+)$$

$f(x)$ is not differentiable at $x = 1$

b) $x = \frac{t}{2t+1}$

$$\frac{dx}{dt} = \frac{(2t+1)(1) - (t)(2)}{(2t+1)^2}$$

$$= \frac{2t+1-2t}{(2t+1)^2}$$

$$= \frac{1}{(2t+1)^2}$$

$$y = (2t+1)^{-3}$$

$$\frac{dy}{dt} = -3(2t+1)^{-4}(2)$$

$$= -\frac{6}{(2t+1)^4}$$

$$\frac{dy}{dx} = \frac{-\frac{6}{(2t+1)^4}}{\frac{1}{(2t+1)^2}}$$

$$= -\frac{6}{(2t+1)^2}$$

c) $y = e^x \ln(1+x)$

$$\frac{dy}{dx} = e^x \left(\frac{1}{1+x} \right) + [\ln(1+x)]e^x$$

$$\frac{dy}{dx} = \frac{e^x}{1+x} + y$$

$$\frac{d^2y}{dx^2} = \frac{(1+x)e^x - e^x(1)}{(1+x)^2} + \frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} = \frac{e^x + xe^x - e^x}{(1+x)^2}$$

$$(x+1)^2 \left(\frac{d^2y}{dx^2} - \frac{dy}{dx} \right) = xe^x$$

7 a) $2\pi rh + \pi r^2 = 300\pi$

$$2\pi rh = 300\pi - \pi r^2$$

$$h = \frac{150}{r} - \frac{r}{2}$$

$$V = \pi r^2 h$$

$$= \pi r^2 \left(\frac{150}{r} - \frac{r}{2} \right)$$

$$V = 150\pi r - \frac{\pi r^3}{2}$$

b) $\frac{dV}{dr} = 150\pi - \frac{3\pi r^2}{2}$

when $\frac{dV}{dr} = 0$

$$150\pi - \frac{3\pi r^2}{2} = 0$$

$$-\frac{3\pi r^2}{2} = -150\pi$$

$$r^2 = 100$$

$$r = 10 \text{ since } r > 0$$

$$\frac{d^2V}{dr^2} = -3\pi r$$

$$\begin{aligned}\text{when } r = 10, \frac{d^2V}{dr^2} &= -3\pi(10) \\ &= -30\pi \\ &< 0 \text{ (max)}\end{aligned}$$

$$\begin{aligned}V &= 150\pi(10) - \frac{\pi(10)^3}{2} \\ &= 1000\pi \text{ cm}^3\end{aligned}$$